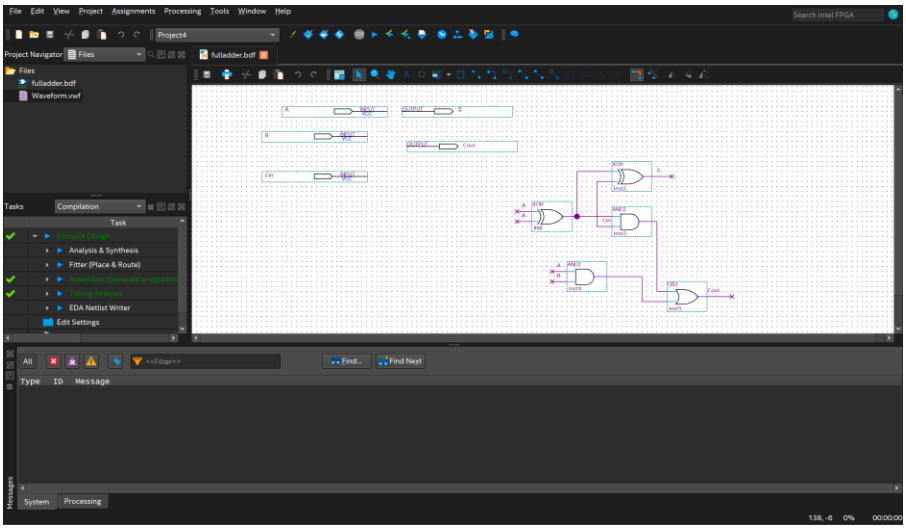
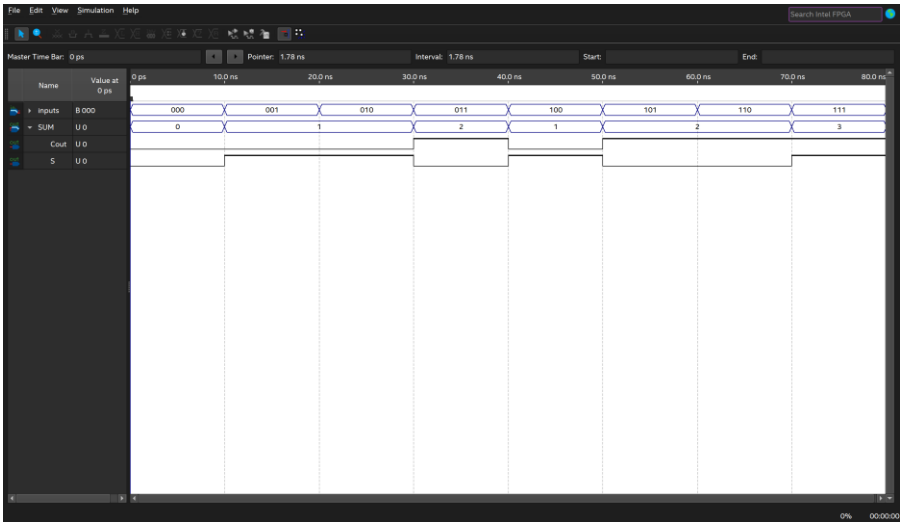
Name: Gazi Daniyal **Date:** 11/8/2025

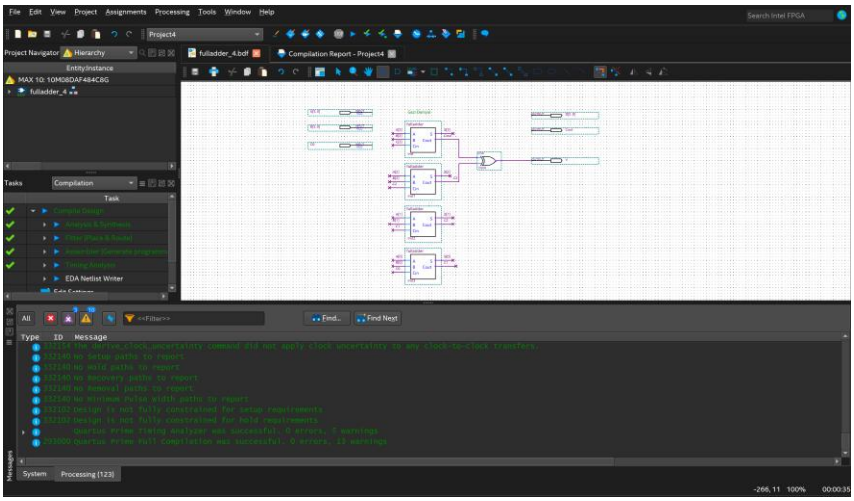
Project 4

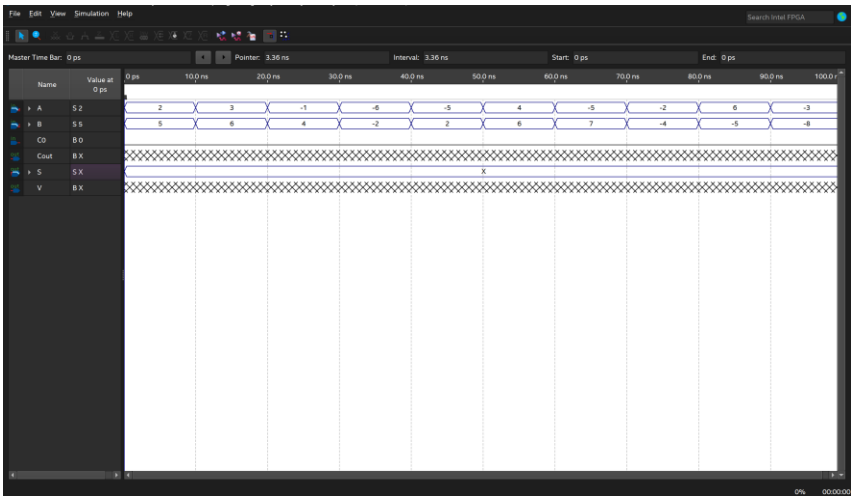
Part I: Designing a Full Adder

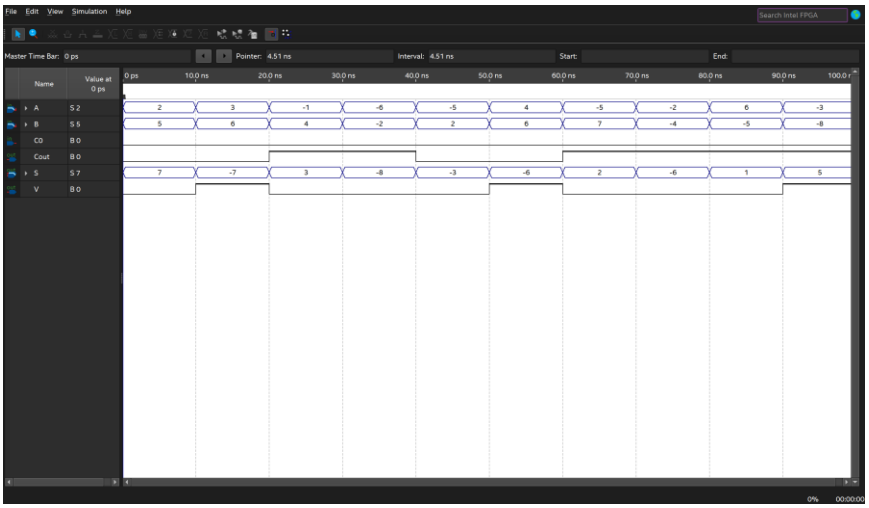




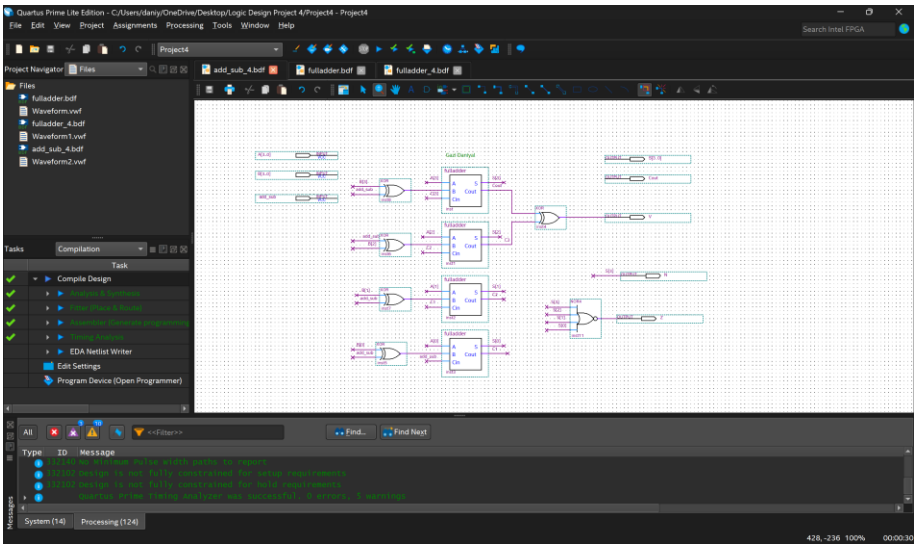
Part II: Hierarchical Design

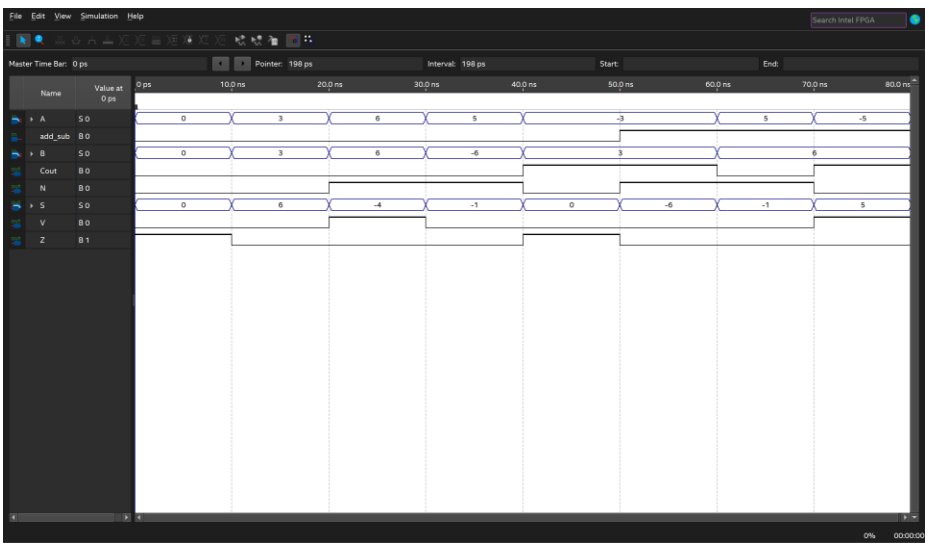




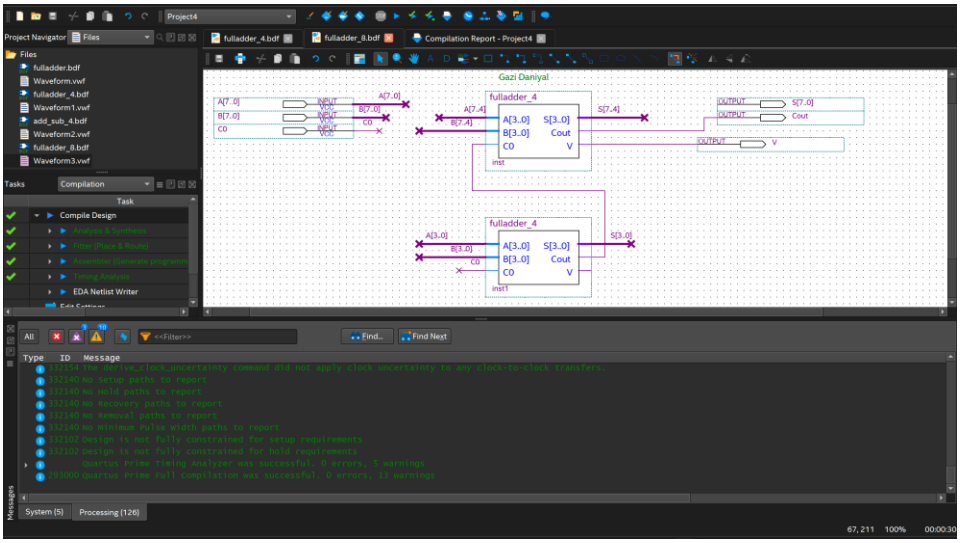
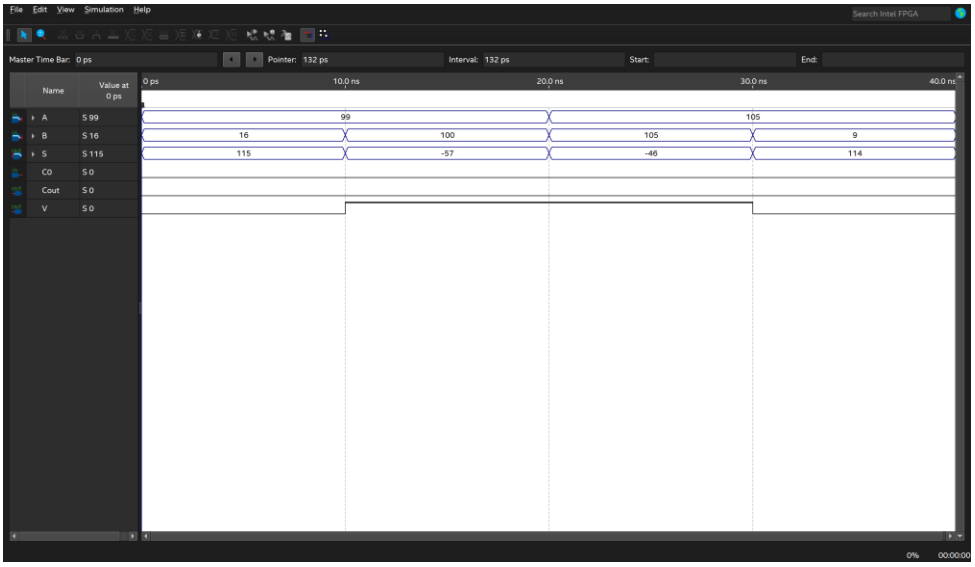


Part III: The Adder/Subtractor with additional Flags





Description:

The purpose of part3 is that it shows a 4-bit adder-subtractor circuit that can add and subtract in response to a control signal that is demonstrated on the circuit. Which generates carry and overflow outputs, a 4-bit result, and two 4-bit binary inputs. And then the waveform shows how the circuit should work by generating precise results for a range of input combinations during addition and subtraction operations.

Part IV: 8-bit adder